

Engineered wood - it's all in the glue

There are many types of engineered wood, made up of wood chips, veneers or strands bonded together with petrochemical resins (like urea-formaldehyde).

The most common types

 MDF Medium Density Fibreboard Stage sets, cabinetry & panelling	 Particleboard/ Chipboard Set elements, signage, non-structural	 OSB Oriented Strand Board Flooring, platforms & panelling	 LVL Laminated Veneer Lumber Formwork, beams	 Plywood & Formply Cross Laminated Veneers Robust temporary platforms, sub floors, hoardings
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Concerns



Chemical resins

Urea-formaldehyde (UF), a known carcinogen, is released over its lifecycle.



Non-renewable

Resins are typically petroleum based, dependent on fossil fuels.



Difficult disposal

Often cannot be recycled because of glues and chemical treatments, making it destined for landfill.



Golden rule

Unless explicitly labelled low-formaldehyde/ ULEF/ NAF/ Super E0 - **assume** engineered wood products contain some formaldehyde.

Read the glue: what to avoid vs what to choose

← Toxic formaldehyde resins

Less + natural is better →

E2	E1	MUF	NAF	PUR	PF	E0	Super E0 / ULEF	NAF	PVA
Higher emission < 2.0 mg/L formaldehyde	Lower emission ≤ 1.0 mg/L formaldehyde	Melamine-Urea-Formaldehyde	No Added Formaldehyde - PMDI (polymeric MDI) petrochemical	Polyurethane - oil-based, not biodegradable	Phenol-Formaldehyde resin, petrochemical	Very low emission ≤ 0.5 mg/L formaldehyde	Ultra-low emission ≤ 0.3 mg/L formaldehyde	No Added Formaldehyde Bio-based binders	Polyvinyl Acetate Adhesives

Avoid

Toxic, fossil fuel based, products non-recyclable

Choose

Responsible Materials that have super low petrochemical resins and can be reused at their end of life or recycled.

Look for these standards

Joinery, cabinetry & wall flats

MDF is the default for joinery, but its resins make it almost impossible to recycle. These engineered-wood alternatives are price-competitive and recyclable when finished right.

Alternatives to MDF

PRODUCT	SUPPLIER	PRICE	LOOK FOR	CIRCULAR (RECYCLABLE/ REUSABLE)
Keep using: already at the AO				
Laminex Decorated Particleboard (Low Pressure Melamine)	Laminex	18mm \$30/m ²	E0 or Super E0	Recyclable. Choose particleboard substrate
Laminex Raw Particleboard (Trade Essentials®)	Laminex	18mm \$12/m ²	Particleboard substrate	Recyclable
Structural ply (Type A-bond) & FormPly	Big River Group Auswood Evolution FP	19mm \$75 · \$30–\$37/m ²	PF / A-bond resin	Recyclable
Durra Panel wall, ceiling & acoustic systems	Durra Panel	Price competitive	Natural resin	Recyclable
Responsible alternatives				
EGGER Eurodekor® MFC panels, Super E0	Forest One	18mm from \$40/m ²	E0 or Super E0	Recyclable. 76% recycled content. EGGER closed-loop chipboard.
EGGER PerfectSense® lacquered particleboard	Forest One	18mm \$74/m ²	Particleboard substrate	Recyclable. 68% recycled EGGER closed-loop
EGGER OSB 3 E0 oriented strand board	Forest One	18mm \$89/sheet	Strandboard substrate	Recyclable
SELEX® plywood structural B/D	Forest One	18mm \$80	Particleboard substrate	Recyclable
Long-life: for evergreen elements reused 3+ years				
Melamine particleboard substrate (white)	Big River Group Auswood Evolution FP	16mm \$60 · 18mm \$80	PF / A-bond resin	Some recycling supported Source melamine with chipboard substrate
Laminex Compact Laminate (phenolic core)	Laminex	\$140–\$185/m ²	Long-term asset	Difficult to recycle : landfill or waste-to-energy
Paperock SOLID phenolic paper composite	Paperock	\$110/m ²	Long-term asset	Non recyclable. Excellent durability - long life reduces replacement frequency
AA grade marine ply BS1088	Auswood	15–25mm \$90–\$150	MUF / UF resin	While it is a sound material for reuse it cannot be recycled. Consider alternatives: F17 Radiata Pine ply/ formply

AO Target Materials List · prices indicative

Methods matter

- **Fix mechanically:** screws, bolts, knock-down fittings or [Button-fix](#), not glue.
- **Water-based paint only** ([Wattyl ID](#), [Dulux Enviro](#), [Paintback](#)).
- **Graphics that come off:** removable SAV; ACP that separates from ply.
- **Build modules** sized for storage, and document how they come apart.

Limit use

- **Non-structural CD ply:** Resin is MUF/ UF cannot be recycled (waste stream)
 - **MDF (décor/ MR, raw):** Bonded with toxic MUF/ UF which cannot be recycled (waste stream)
Consider alternatives: Melamine/ Formica/ Compact Laminate
- If you must use MDF, use in modular evergreen, elements built for multi-year reuse over single use.**

TA Fire/Safety

Cabinetry and wall flats permanently fixed to floors or walls require a **Group 1 or 2** rating with a minimum **Critical Radiant Flux of 2.2 kW/m²**.

Ask for AS 1530.3 test reports first; if unavailable, AS 5637.1.

Specifications: Engineered wood & alternatives

Typical builder choices, their resin composition and their waste fate

PRODUCT	WHAT IT IS	TYPICAL RESIN	AT END OF LIFE
1. Panel products			
Plywood	Veneers bonded in alternating grain layers for strength and dimensional stability; used in structural sheathing, flooring and cabinetry. Excluding Marine ply which cannot be recycled.	PF (exterior), MUF or MUF (interior)	Recyclable
Formply / Formwork Plywood	Structural plywood with phenolic film overlay designed for concrete formwork and repeated wet exposure.	PF (phenol formaldehyde)	Recyclable
Particleboard / Chipboard	Small wood particles bonded with resin and pressed into sheets used for furniture cores and laminate substrates.	UF (most common), PF for moisture-resistant grades	Recyclable
Melamine-Faced Panels	Particleboard or MDF core panels coated with decorative paper impregnated with melamine resin.	Melamine-formaldehyde surface + UF core resin	Recyclable
OSB (Oriented Strand Board)	Large oriented wood strands pressed in cross-layers for structural panels used in walls, floors and roofs.	pMDI or PF	Limited / depends
MDF (Medium Density Fibreboard)	Wood fibres mixed with wax and resin and hot-pressed into smooth panels for joinery, cabinetry and interior surfaces.	UF or MUF	Not recyclable
HDF / Hardboard	Very dense fibreboard produced by compressing wood fibres under heat and pressure into thin strong sheets.	UF, PF or natural lignin bonding (wet-process boards)	Limited / depends
LDF (Low Density Fibreboard)	Lower density fibreboard panels used for insulation boards or low-load interior components.	UF	Not recyclable
2. Structural Composite Lumber (SCL)			
LVL (Laminated Veneer Lumber)	Veneer layers bonded with grain parallel to form high-strength beams, lintels and rim boards.	PF or pMDI	Limited / depends
LSL (Laminated Strand Lumber)	Shorter wood strands bonded into billets used for studs, headers and rim boards.	pMDI	Limited / depends
I-Joists	Engineered floor joists using LVL/solid timber flanges with OSB or plywood web panels.	PF or pMDI in web/flange bonding	Limited / depends
3. Glued Laminated & Mass Timber			
Glulam (GLT)	Structural beams or columns made by bonding timber laminations with structural adhesives.	PRF, PUR, MUF or EPI	Reuse
CLT (Cross-Laminated Timber)	Thick multi-layer timber panels with cross-oriented boards used for walls, floors and roofs.	PUR, PRF or MUF	Reuse
NLT / DLT	Solid timber boards mechanically fastened together with nails or dowels to form floor or roof panels.	None (mechanical fastening)	Reuse
Mass Plywood / MPP	Thick structural panels made from stacked veneer layers similar to plywood for mass-timber construction.	PF	Reuse
4. Alternatives: decorative laminated & composite panels			
HPL (High Pressure Laminate)	Decorative laminate made from resin-impregnated kraft papers pressed under high heat and pressure.	Phenolic core + melamine surface	Not recyclable
Compact Laminate	Thick self-supporting HPL panels used for facades, wet areas and high-durability applications.	Phenolic resin	Not recyclable
Wood-Plastic Composite (WPC)	Wood fibre combined with thermoplastics and extruded into decking or exterior trim boards.	PE, PP or PVC polymers	Limited / depends
ACP / ACM (Aluminium Composite Panels)	Aluminium skins bonded to polymer core used for facade cladding and signage panels.	Polyethylene (PE) or mineral-filled polymer core	Recyclable